

Predictive Analysis of Health Risk Associated with Delayed Health Care Seeking Behavior Using Machine Learning

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Abstract: Postponement of seeking medical care after the symptom onset plays a role in advancement of disease, extended hospital stays, increasing treatment expenses and avoidable deaths. The low level of financial strain, geographic distance, limited health literacy and underestimation of the seriousness of their own symptoms can contribute to avoidance in clinical settings. Most existing work in this area only models the predicted probability of a delay occurring rather than how much health risk is generated by that delay. This paper introduces a hybrid deep learning architecture in which an Autoencoder extracts latent structural patterns from patient records and feeds the resulting compressed representation into an Artificial Neural Network regression model, which then produces a continuous numeric health risk score. While working with public dataset set it was noticed that multiple variables like socioeconomic background, period of delay, severity of observed symptom, geographical distance from their location were absent. A synthetic dataset which included 6000 samples was created based on the patterns and findings observed in the peer review studies related to healthcare delay. To focus mainly on working age class whose age lies between 18 to 60 the samples were filtered and final 3836 was taken because this group lacks taking medical advice due to busy life schedules. Before training the data it was prepared by filling missing values with mean and modes, converted categorical values like income, insurance status into numerical form using one-hot encoding. Z-score normalization was used to standardized all continuous variables. This was done to avoid single variable dominate the training process simply because of its scale. To capture hidden feature a bottleneck Autoencoder was trained. The compressed output was then combined with original preprocessed features to form hybrid input for ANN regression model. The model achieved an MAE of 4.19, RMSE of 5.18 and R^2 score with 0.94.

Keywords: *Delayed Health Care, Health Risk Prediction, Autoencoder, Artificial Neural Network, Latent Feature Extraction, Predictive Analysis, Risk Score Regression.*

I. INTRODUCTION

Getting help on time plays key role in saving one's life and also helps to keep medical costs down. One individual may avoid a clinic because the nearest facility is forty kilometres away or they may ignore symptoms entirely because the price of losing wages seems much more pressing than the risk of an illness. Lack of trust in doctors, no medical coverage, low symptom awareness and hectic work schedules all contribute to delay. Such negligence can lead to heart attacks, stroke, diabetes, respiratory failure and cancer which have become very

common today. A simple treatable condition when ignored on time can turn life threatening. Surveys and clinical interviews help identify factors behind delay but fail to predict real time risk since they depend on basic assumptions and cannot handle multiple factors together. Machine learning and deep learning can address this by finding patterns across many variables at once for accurate health risk prediction. Most existing models only answer yes or no for delayed care while very few have tried to measure how serious the risk actually is. The main focus of this paper is to consider severity as continuous quantity. This paper builds a system by combining an Autoencoder for unsupervised latent feature extraction with an ANN regression model to produce a numeric severity score.

II. LITERATURE SURVEY

Published work on healthcare seeking delay spans a wide range of methodological approaches and disease contexts, yet nearly all of it frames the problem as binary classification. Using hybrid deep learning to quantify continuous health risk severity from delay related variables has received little attention.

Chen, et al. examined hospital records from breast cancer patients and used Support Vector Machine classifiers, logistic regression, Random Forest and XGBoost to predict delayed care seeking. Tree based models outperformed logistic regression, with Random Forest and XGBoost producing stronger results. Validation accuracy reached 79%. The most influential predictors included low health awareness, financial barriers, rural location, and restricted facility access. The study did not attempt to quantify how severe the resulting health risk was from any given delay. [1]

Sheng, et al. studied elderly stroke patients and used LightGBM alongside Random Forest and logistic regression to predict which patients would delay seeking emergency care. LightGBM reached 80.6% accuracy, outperforming the other two models. The strongest predictors of delay were mild symptom perception and insufficient awareness of stroke signs. The study was disease specific and did not move beyond classifying whether delay occurred. [2]

Yang, et al. conducted a bibliometric analysis of artificial intelligence applications in healthcare risk prediction, drawing from Web of Science and PubMed. Their findings indicate that AI based risk prediction is concentrated almost entirely in disease diagnosis and treatment outcome modelling. Predicting risk severity arising from delayed care seeking was identified as an underexplored area. [3]

Davy, et al. used electronic health records to study delayed discharge rather than care seeking delay at the patient level.

Logistic regression and statistical analysis identified care coordination failures and administrative bottlenecks as primary contributors to discharge delay, achieving 70% classification accuracy. Deep learning method were not incorporated. [4]

Frosch, et al. developed a multilevel predictive model using cancer registry and hospital administrative data to identify patients at risk of delayed treatment initiation. Patient, provider, and system level factors were all included. Group LASSO achieved an AUC-ROC of 71.3%. Estimating risk severity was outside the scope of that work. [5]

Van Gaal, et al. examined carotid artery stenosis treatment delays using insurance claims data. Regression modeling, clustering, and waiting time analysis were combined to study what slows care delivery. Clustering effectively separated fast and slow treatment pathways. No continuous predictive risk scores were generated. [6]

Wang, et al. applied Random Forest, XGBoost, ANN, and SVM to predict prolonged emergency department wait times. Explainable AI techniques were used to interpret model outputs. Accuracy across models ranged from 74% to 75%. The research focus was on hospital operational management rather than patient side delay behaviour. [7]

Deina, et al. used decision trees, Random Forest, and gradient boosting to predict appointment nonattendance from scheduling records. Insurance status and prior attendance history ranked among the strongest predictors. The study was limited to scheduling behaviour and did not address downstream health consequences. [8]

Gertz, et al. investigated delayed care seeking in the United States during the COVID-19 pandemic using survey data and regression analysis. Late access was linked with deteriorating health status and increased hospitalization rates. Primary causes included fear of infection, financial stress, and system overcrowding. No predictive model was developed. [9]

Nab, et al. surveyed emergency department visitors in the Netherlands during the first COVID-19 wave. Misinterpretation of symptoms and concern about overwhelming hospital staff were widely reported reasons for delay. The study was qualitative in orientation and did not include model performance metrics. [10]

Soh, et al. used logistic regression on breast clinic survey data to identify predictors of delayed presentation for breast symptoms. Financial barriers and misunderstanding of symptom significance were the most consistent factors. The model reached 80.2% accuracy with an AUC of 67.7%, but it was restricted to a single disease domain. [11]

Rzeszutek, applied behavioural economics modelling to analyse how symptom severity and out of pocket cost interact in healthcare decision making. Socioeconomic and behavioural factors were found to drive delay particularly when no symptoms appeared severe. The research was theoretical and lacked to report machine learning metrics. [12]

Taiwo, et al. developed an analytic workflow to utilize predictive analytics for reducing diagnostic delay in chronic illness. Accessibility, behavioural, and socioeconomic dimensions were included in the framework but it had not been implemented or tested. [13]

Coombs, et al. conducted interviews with rural healthcare providers to document access barriers. Transportation difficulties, facility shortages, and limited health literacy dominated the findings. The study was qualitative. [14]

Petrova, et al. analysed how varying symptom profiles in acute coronary syndrome affect prehospital delay among elderly patients. Mild or atypical symptom presentation consistently led to longer delays in seeking care. Deep learning and continuous risk modelling were not considered. [15]

Smith, et al. studied delay and care avoidance behaviour among insured adults using survey data. Financial strain and limited understanding of insurance coverage contributed to delay even among people who technically had insurance, demonstrating that being insured alone does not lead to timely care seeking. [16]

Taken together, the reviewed literature confirms that lag has been widely reviewed as a classification task. What remains absent is a continuous, severity based health risk score derived from the combined socioeconomic, behavioural, and clinical cause for delay and a hybrid deep learning architecture capable of capturing the nonlinear structure that governs those relationships. This paper tackles both gaps.

III. METHODOLOGY

A. System Overview

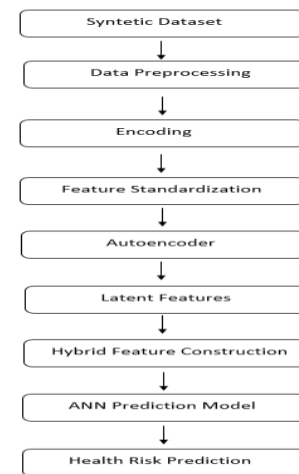


Fig. 1. Overall System Architecture Diagram

The proposed system estimates a continuous health risk rating for people who have hesitated seeking medical attention. Instead of collapsing the problem into a binary result, it generates a numeric severity score that communicates gradations of risk across the entire spectrum. This pipeline works as follows: Dataset collection and validation, Preprocessing and data cleaning, One hot encoding of categorical features, Feature standardization using

StandardScaler, Exploratory data analysis & correlation examination, Latent feature extraction via Autoencoder, Hybrid feature vector construction, Health risk score prediction via ANN regression, Risk category classification from predicted scores.

B. Data Collection and Dataset Design

No existing public dataset combines the specific set of attributes essential to study symptom severity, delay duration, education level, financial status, insurance coverage, geographic distance to a facility, and health risk outcome within a single source. A synthetic dataset was therefore constructed using deterministic rule based logic anchored in findings from published studies on cancer treatment delay, stroke care timing, healthcare access barriers, and utilization behavior among working age populations. The final dataset contains 3,836 records after filtering to include only people with age 18 to 60. This range was chosen as healthcare delay in this group is most strongly shaped by the financial and behavioral variables that the model captures. The data was arranged into training and testing sets using stratified 80:20 split thus maintaining the proportional representation of classes in train and test datasets.

Variables included in the dataset:

- Age
- Gender
- Income Level
- Education Level
- Insurance Status
- Symptom Severity Level
- Healthcare Delay Duration (Days)
- Distance to Healthcare Facility

The risk measure was calculated based on the following logic, grounded in the delay literature: elevated symptom severity combined with extended delay duration produces higher risk scores; absence of insurance and lower income amplify the likelihood of delay and compound outcome severity; greater distance to a facility worsens access and further increases risk. These relationships are monotonic and interactional. In particular, the target score was calculated as follows:

Risk score = $0.35 \times \text{Symptom severity} + 0.30 \times \text{Delay Duration normalized} + 0.15 \times \text{Distance normalized} + 0.10 \times \text{Income penalty} + 0.10 - \text{Insurance penalty}$

where penalty terms are higher for lower income levels and those without insurance coverage. Before combining, all component values were scaled to a 0-100 range.

C. Data Preprocessing

Imputation of missing values: Numerical columns were filled by using the column mean. The proposed work imputed categorical columns with the column mode.

One hot encoding: Gender, Residence, Education Level, Income Level and Insurance Status were encoded by Pandas `get_dummies()` with `drop_first=True` option which removes one redundant dummy column per variable to avoid the problem of perfect multicollinearity in feature matrix.

Feature standardization: The Z-score transformation was applied to all the features with Scikit-learn's StandardScaler.

$$z = (x - \mu) / \sigma$$

where x is the raw feature value, μ is the column mean, and σ is the standard deviation. This step guarantees that features of different scales do not dominate the effect of training with respect to gradients during neural network training.

Exploratory analysis: Histograms, scatter plots, and correlation heat maps were generated to examine feature distributions and pairwise relationships. Symptom severity, delay duration, and distance to facility showed the strongest positive correlations with health risk score, which is consistent with the rules used to construct the dataset and with the patterns documented in the delay literature.

D. Latent Feature Extraction Using Autoencoder

An Autoencoder is a neural network trained to compress its input into a lower dimensional representation and then reconstruct the original input from that compressed form. Because the bottleneck layer has fewer dimensions than the input, the encoder is forced to retain the most structurally informative content including nonlinear dependencies among variables that standard scaling procedures cannot surface.

Encoding step:

$$z = f(Wx + b)$$

where x refers to input vector, W is the weight matrix, b is the bias term, f is the ReLU activation function, and z is the resulting latent representation. The formulation for encoding and decoding was implemented using TensorFlow/Keras-based deep learning architecture.

Decoding step:

$$\hat{x} = g(W'z + b')$$

The Autoencoder was trained using mean squared error reconstruction loss and the Adam optimizer over 50 epochs with a batch size of 16. After training concluded, only the encoder portion was retained. It was applied separately to the training and testing sets to extract a four dimensional latent vector for each record.

E. Hybrid Feature Construction

The latent output produced by the encoder was concatenated with the original standardized feature vector for each sample. This creates a hybrid input that combines direct feature signals with the compressed structural patterns identified by the Autoencoder. The ANN regression model trains on this combined representation, not on either component independently. The benefit of this approach is that the regressor receives both the interpretable input dimensions and the hidden relational structure that the unsupervised component uncovered.

F. ANN Regression Model

The hybrid feature set was passed into a feedforward ANN implemented using TensorFlow Keras. Dense layers to predict continuous risk scores. The neuron computation performed at each layer follows:

$y = \sigma(\sum w_i x_i + b)$

where x_i are input features, w_i are learned weights, b is the bias, σ is the activation function, and y is predicted risk value.

G. Training Configuration

Training and validation loss were tracked across all 100 epochs to monitor generalization behavior. The final training loss of 3.98 and validation loss of 4.21 indicate that the difference between both of them remained narrow throughout, reflecting stable learning without evidence of significant overfitting.

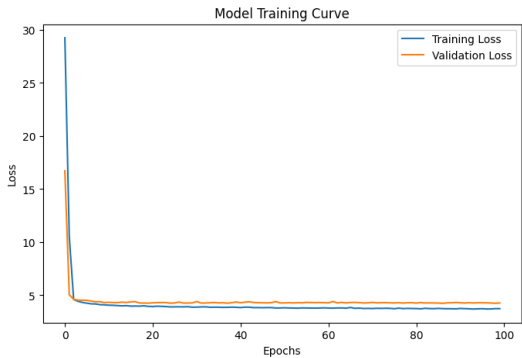


Fig. 2. Training and Validation Loss Curve

Table I: Training Parameters

Parameter	Value
Autoencoder optimizer	Adam
Autoencoder loss	MSE
Autoencoder epochs	50
ANN optimizer	Adam
ANN loss function	MAE
ANN epochs	100
Batch size	16
Validation split	20% of training set
Train-test split	80:20 stratified

H. Risk Score Prediction and Categorization

Thresholds were set at 50 and 75 based on the synthetic score distribution. The lower threshold separates patients where delay has produced limited projected deterioration from those showing moderate compounded risk. The upper threshold marks the point at which symptom severity and delay duration jointly indicate a high probability of serious clinical consequence. Class balance across the three categories was verified on the test set before finalizing these values. Predicted continuous scores were mapped to three interpretable risk categories as follows:

Table II: Risk Categories

Risk Score	Category
Score < 50	Low Risk
50 ≤ Score < 75	Moderate Risk
Score ≥ 75	High Risk

IV. RESULTS AND EVALUATION

1. Regression Performance

An MAE of 4.19 indicates that the average deviation between predicted and actual risk scores is under five points on the full

0-100 scale. An R^2 value of 0.94 means the model accounts for 94% of the variance in health risk scores across the test set a strong result for a regression task of this nature.

Table III: Model Results

Metric	Value
Mean Absolute Error (MAE)	4.19
Root Mean Squared Error (RMSE)	5.18
R^2 Score	0.94
Final Training Loss	3.98
Final Validation Loss	4.21
Epochs	100
Batch Size	16
Optimizer	Adam

2. Learning Curve

LEarning and validation loss decreased consistently across all 100 epochs without observable divergence. The narrow final gap between training loss (3.98) and validation loss (4.21) confirms that the model generalized effectively to unseen records rather than memorizing patterns from the training set.

3. Classification Performance

Mapping the continuous regression output to the three risk categories produced an overall classification accuracy of 94.14%. ROC analysis using a One-Vs-Rest strategy yielded the following AUC values:

- Low Risk: AUC = 0.99
- Moderate Risk: AUC = 0.96
- High Risk: AUC = 0.99

These results confirm strong discrimination across all three severity levels, not merely at the aggregate level.

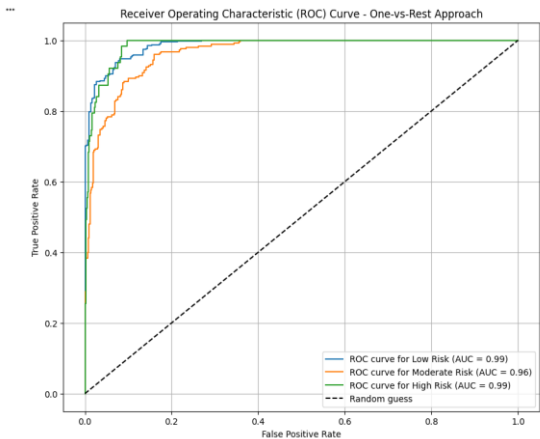


Fig. 3. ROC Curve Analysis for Healthcare Risk Categories

4. Correlation Analysis

Symptom severity and delay duration showed the strongest positive correlations with the health risk score. Distance to the nearest facility followed as the next most influential variable. Income level and insurance status both demonstrated moderate but consistent effects on delay duration and on the final risk score. This pattern aligns with what the delay literature predicts and is consistent with the logic embedded in the dataset construction process.

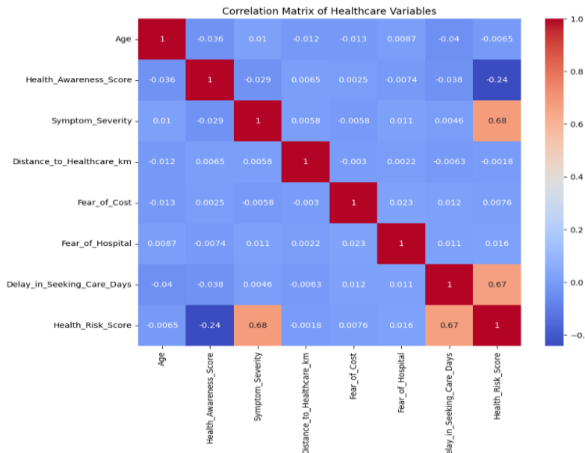


Fig. 4. Correlation Heatmap Showing Relationships Between Healthcare Variables

5. Prediction Scatter and Error Distribution

Actual versus predicted scores were tightly clustered around the diagonal line of perfect agreement, with most points falling in close proximity to it. The distribution of prediction errors was approximately normal and centered near zero, with no systematic directional bias in either over- or under-prediction. This confirms that the hybrid feature approach improved the ANN's approximation of the underlying risk function.

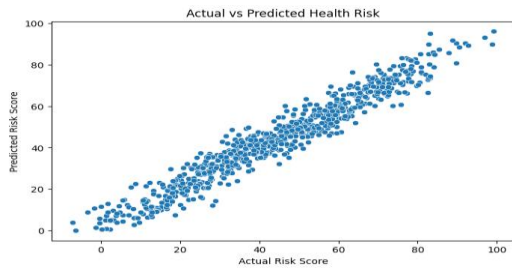


Fig. 5. Actual vs Predicted Health Risk Scores

6. Error Distribution Analysis

Prediction error analysis was carried out to assess consistency across the test set. The concentration of errors near zero indicates low average deviation and confirms that the hybrid feature construction approach meaningfully enhanced predictive performance. The absence of a systematic directional skew in the error distribution further supports the stability of the model.

7. Feature Influence

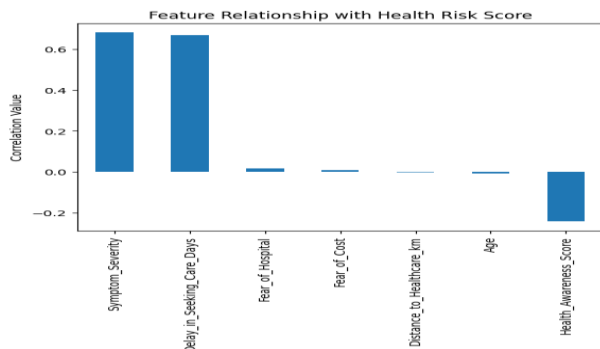


Fig. 6. Feature Influence Analysis

Permutation importance analysis identified delay duration and severity of symptom as the two most influential predictors, followed by distance to a healthcare facility. Notably, the latent features contributed meaningfully alongside the original inputs, confirming that the Autoencoder extracted genuinely informative hidden structure rather than producing redundant representations of what the raw features already captured.

8. Prediction Interface

An interactive healthcare risk prediction interface was developed to demonstrate real-time applicability. Users enter demographic and clinical data including age, symptom severity, delay duration, income level, and insurance status and the trained ANN returns a predicted risk score along with a risk category label and an advisory message. This interface was built using a Python-based web framework and is intended as a demonstration of how the model could be deployed in community health or mobile health settings.

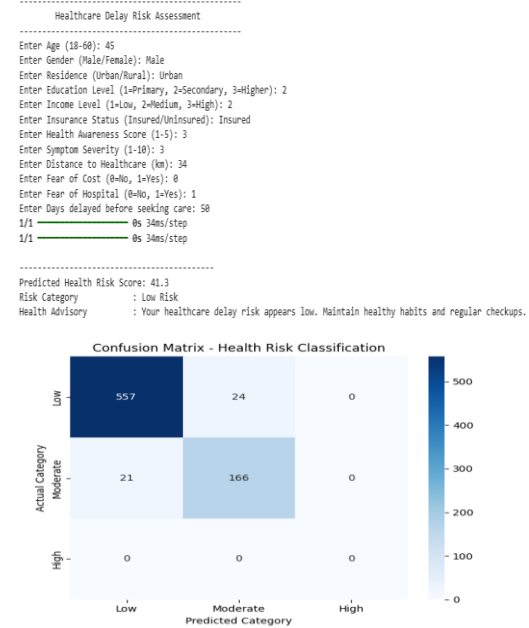


Fig. 7. Healthcare Risk Prediction Interface

V. DISCUSSION

This study shows hybrid risk prediction based on latent feature learning using Autoencoders combined with regression using ANNs yields greater predictive power compared to conventional binary risk prediction based on classification algorithms. Several issues raised by these results are worth discussing further. Healthcare delay is not explained by just one factor. The analysis shows that health condition severity, delay days, distance to healthcare facilities, income level, presence of health insurance, and education are factors that explain healthcare risk, and the interactions among these variables have an effect that goes beyond each individual variable alone. The Autoencoder learns this interaction by attempting to recreate the entire input space through a compressed bottleneck, encoding the interaction of the features as its output and enabling the ANN to leverage that output to learn its regression task. Major change is using prediction framework as regression rather than classification. Because classification results are either true or false, which hides the fact that actually two

persons are at different risk level. The proposed model gives number which consists of important information for people to take right decision. To make it easier to access a simple prediction interface was built where one can enter personal details and get their result. User can enter age, gender, distance to healthunit, awareness level, income level, insured or not, symptom impact and get resultant score. Constraint of the study is that data was synthetic, but is was made using published studies. Patients choice depends on emotions, social influence and cultural practice. Therefore before make use of this model it has to be tested using real patient data. Along with this, it will be good to compare it with basic ANN and Random forest model inorder to understand contribution of used hybrid model.

VI. CONCLUSION AND FUTURE SCOPE

This study used a hybrid structure combining Autoencoder and ANN regression to predict risk score for patients who delay treatment. Autoencoder pulled out hidden patterns from the data while ANN used those patterns along with original features to generate the final score. When tested on 3836 records it gave MAE of 4.19, RMSE of 5.18 and R^2 of 0.94. Getting R^2 of 0.94 shows that predicting health risk as a continuous score gives much more useful information than a simple yes or no label. The only limitation of this study is that synthetic data was used. Validation against real hospital records or EHR datasets is a prerequisite before clinical deployment. Future directions include: validation against actual EHR and hospital records; integration of real-time physiological signals from wearable devices and IoT health monitors; incorporation of SHAP-based explain ability to improve model interpretability for clinical practitioners; exploration of LSTM-based sequence models for capturing temporal symptom progression over time; and deployment as a web or mobile risk assessment tool targeting underserved populations with limited facility access.

VII. REFERENCES

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